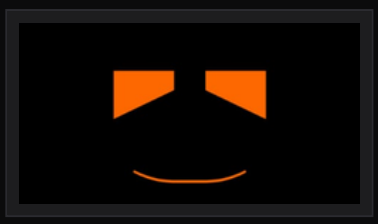
AUTONOMOUS COMPANION ROBOT · RASPBERRY PI · BUILT BY HARSH

FRIDAY

Hi, I'm **Friday**. Harsh built me — the chassis, the wiring and the code. I wake when you say my name, I look before I move, I keep what you teach me, and if I don't know something, I'll just say so.



MY FACE · 7″ SCREEN



FRIDAY'S ACTUAL BODY · 3D-PRINTED



OUT FOR A DRIVE

BODY

- **Raspberry Pi** — the brain box
- **360° LiDAR** — on the roof
- **2 cameras** — room + floor
- **8×8 ToF** — depth at the bumper
- **7" screen** — her face
- **Mic + speaker** — ears and voice
- **4 motors** — L298N drive

20 FACES

She picks the one that fits what she just said — safety faces always win.



01 Brain & Voice

- **"Hey Friday"** wake word, double-checked
- Speech-to-text on the Pi itself
- Natural voice — English, Hindi or Hinglish
- Claude brain with **live web search**
- Knows when the TV is talking, not you
- Drive, stop, go home, explore — **offline**

02 Vision

- Two cameras: the room and the floor
- Object detection, always updating
- Notices something new — and mentions it once
- **"What do you see?"** — she looks first, every time
- Tells you what she saw after exploring

03 Navigation

- Maps the room with a 360° LiDAR
- Finds herself on her saved map
- **"Remember this place" → "go to the kitchen"**
- Go home, come back, try again
- Explores on her own, then **returns to her start**
- Backs off when stuck — and asks for help

04 Assistant & Memory

- Remembers facts between conversations
- **Memory Lane:** what she knows about people
- Alarms, timers, clock & weather on her screen
- Phone dashboard: cameras, maps, logs
- Daily spend cap, so she can't run up a bill

05 Self-Improvement

- Correct her once — she keeps it
- Notes what she didn't know...
- ...and looks it up quietly when idle
- Reflects on her day; **"how much have you grown?"**
- Edits her own code — tested, safety locked

06 Study Companion

- Drop in your PDF notes
- Ask questions — answered from **your** notes
- Flashcards with spaced repetition (FSRS)
- Knows which cards are due, and when
- Study mode on the phone dashboard

SAFETY

Every move is guarded by the server, not trusted to the brain · a bumper depth sensor stops her in centimetres · **"stop" always wins** · no wheel encoders, so she says **"about"** and means it · when truly stuck she stops and waits for a person.

> **Before I move, I look. Every single time.**
> — FRIDAY

> **"I don't know" is an answer. A guess is how robots get broken.**
> — FRIDAY

> **Kuch naya dikha? Chalo, dekh ke aate hain.**
> — FRIDAY · saw something new? let's go look

> *The world is its own best model.*
> — RODNEY BROOKS, 1990

57	20	360°	2	8×8	1
TOOLS	FACES	LIDAR	CAMERAS	DEPTH GRID	RASPBERRY PI

BUILT BY HARSH

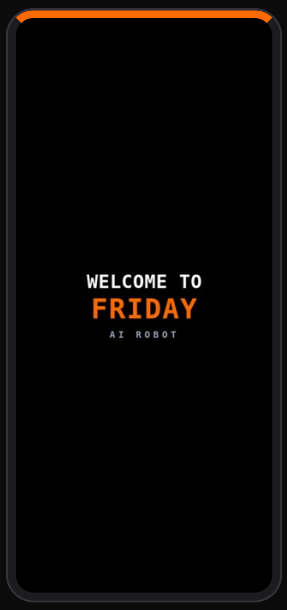
Friday · 2026

HER APP

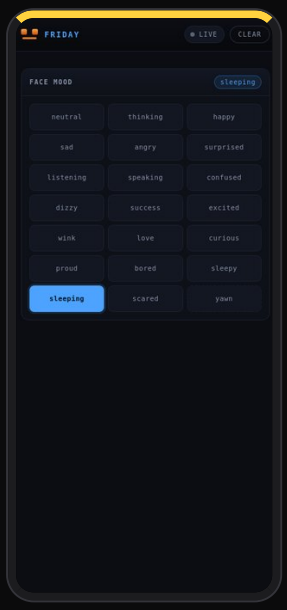
Open me on your phone and you can see what I see, drive me, set an alarm, send me exploring or study with me — and read every thought I had while doing it. No cloud in the middle: it's served straight from me.

INSTALLS LIKE AN APP

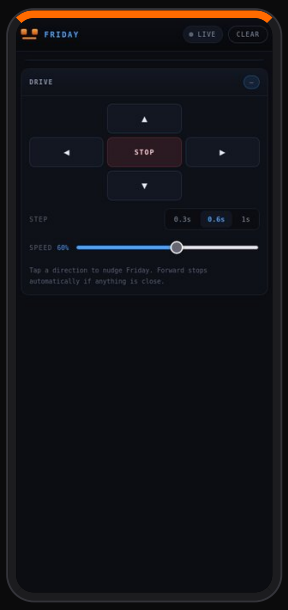
Add to home screen → her orange face is the icon, and it opens with *"Welcome to Friday"*. Runs over the room's Wi-Fi at `https://<friday>:8443`.



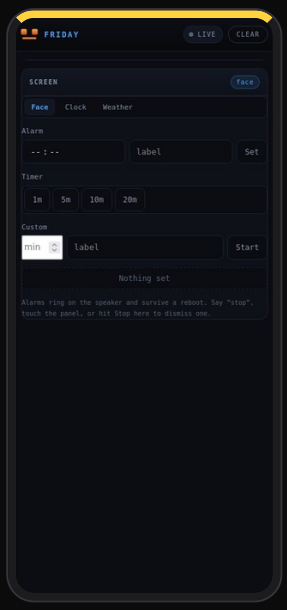
Opens as her
"Welcome to Friday"



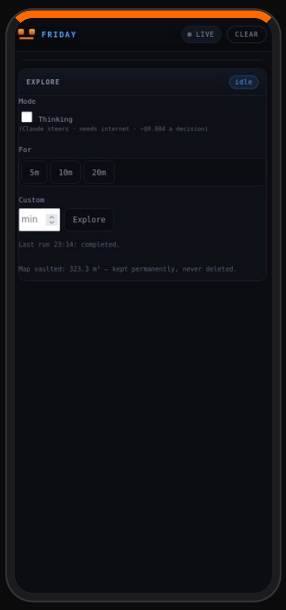
Face Mood
see and set her face



Drive
nudge her, or STOP



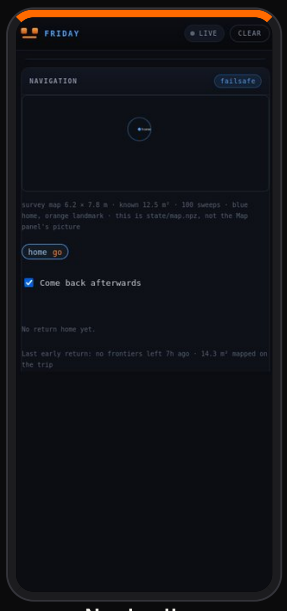
Alarms & timers
ring on her speaker



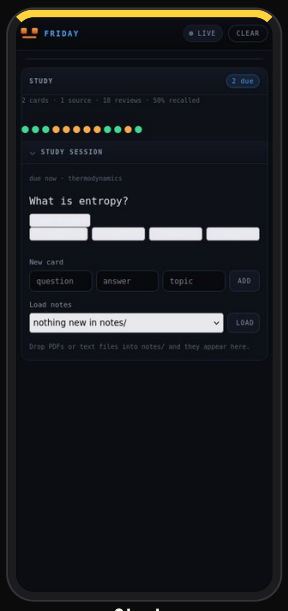
Explore
send her out for N min



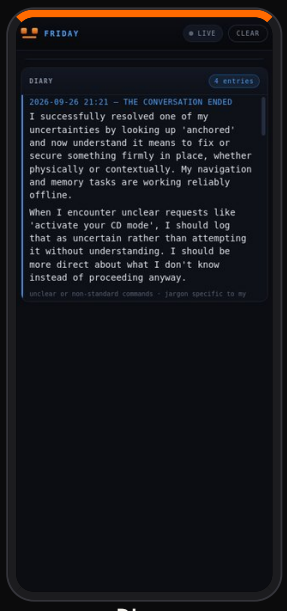
Live map
what she has mapped



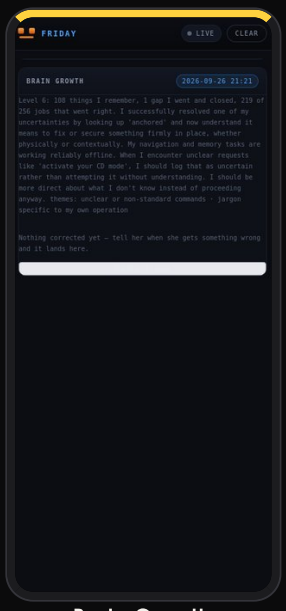
Navigation
places, come back



Study
flashcards from your notes



Diary
her own reflections



Brain Growth
what she's learned

A See & hear

- **Live Logs:** every word she heard, thought, tool and reply
- Up & down cameras, live, switch on when you want
- **Face Mood:** see and set her face
- Says "Robot off" if her body's server stops answering

B Control

- **Drive pad:** forward, back, turn, big **STOP**
- Step size and speed slider
- **Screen:** alarms & timers that ring on her speaker
- **Calibration:** her default move & turn speeds

C Explore & maps

- **Explore** for N minutes, with or without thinking
- Live **Map** as she drives
- **Maps:** every map she ever saved
- **Navigation:** go to a place, come back now, try again

D Learn & grow

- **Study:** load notes, flashcards — Again / Hard / Good / Easy
- **Diary:** her own reflections
- **Memory Lane:** what she knows about people
- **Brain Growth:** what she's learned; "reflect now"

15 PANELS · 1 TAP TO STOP · 0 CLOUD IN BETWEEN